
Cultural-QLoRA: Hofstede-Conditioned Persona Adaptation for Small Korean LLMs

Korea University COSE461 Final Project

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INTERIM PEER-REVIEW DRAFT — 2026-05-29 KST

This draft incorporates fixes from a 2026-05-29 multi-agent audit (Claude sub-agent + Codex-style independent review). Currently being addressed in background AWS runs (ETA ~50 min from this version):

1. **Full $n = 150$ per-question bootstrap CI** for headline cells (currently $n = 20$ subset stored; underlying eval used full 150). Will narrow CIs by $\sim 2.74\times$ and may resolve UAI-only vs. Cultural-KR significance.
2. Multi-seed Cultural-KR results just retrieved from AWS (seeds 42/123/7777; mean $\pm$ std = 0.617 ± 0.025 on KR alignment); integrating into Table 4.
3. Krippendorff's α for CAS panel (replacing mean pairwise diff per Codex critique) — Mac-side compute, ~ 15 min.
4. JP/CN HAE-RAE/CLiCK rerun for Table 3 (currently shown as ‘-’ due to time budget).

Already corrected in this draft (from audit findings): (a) Cultural-KR KoBBQ bias claim flipped from ‘+3.7pp INCREASE’ to ‘-4.5pp DECREASE (smallest among cultural adapters)’ — Abstract & §5 reflect FIXED rerun data; (b) UAI-only “wins” softened to “numerically lower, CIs overlap by 0.07”; (c) Table 6 caption corrected from ‘median’ to ‘mean’; (d) all three `\todo` placeholders resolved with verified prose; (e) 4 silent eval-loader bugs disclosed honestly in §6.

Abstract

Off-the-shelf instruction-tuned LLMs over-represent Anglo cultural priors even when prompted in Korean, limiting their utility for non-Anglo applications such as consumer-persona generation and culturally grounded content. We investigate whether parameter-efficient cultural conditioning—QLoRA fine-tuning on culturally-grounded data (CultureBank, Nemotron-Personas-Korea) with Hofstede-6D system-prompt conditioning—can measurably shift a small (3B/7B) instruction-tuned model’s response distribution toward a target culture’s empirical opinion distribution. We train six cultural adapters (KR, JP, US, CN at 3B; KR at 7B; and a multi-cultural unified variant) on Qwen2.5-Instruct backbones and evaluate against (i) four Korean benchmarks (KoBBQ, KMMLU, HAE-RAE, CLiCK), (ii) cross-cultural alignment via Kolmogorov–Smirnov distance against GlobalOpinionQA empirical distributions plus BLENd MCQ accuracy, and (iii) a 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro) scoring cultural authenticity. Our key finding is that QLoRA-based cultural reweighting on a 3B model produces measurable distributional shifts toward the

target country on multinational survey questions, and transfers across cultures via a single multi-cultural adapter conditioned on a culture token. We also surface an honest asymmetry: while Cultural-JP/US/CN adapters reduce KoBBQ bias by 18–24pp, the Cultural-KR adapter shows the smallest debiasing effect (−4.5pp: 0.450 to 0.405), suggesting that training on target-culture data may amplify in-culture stereotype patterns. We frame this as an alignment–debiasing asymmetry distinct from generic bias mitigation. Code, adapters, decision log, and reproducibility scripts are released.

1 Introduction

Large language models (LLMs) increasingly mediate decisions in consumer modeling, persona generation, and cross-cultural product design. A growing body of evidence shows these models embed predominantly Anglo-American cultural priors [1, 2], which presents a structural problem for non-Anglophone users. Even when prompted in Korean, Vanilla Qwen2.5-7B-Instruct scores only 32% on KMMLU Korean-History and 47% on the HAE-RAE Bench 1.1, while exhibiting a 35% KoBBQ bias rate driven by Anglo-default framings of Korean social contexts.

Prior work either (a) measures the cultural-bias gap without remediation [1, 3, 4] or (b) aligns models via full RLHF on cultural preferences [5, 6], which is impractical for small-team deployments. We investigate a middle path: **parameter-efficient cultural conditioning** via QLoRA [7] with Hofstede-6D [8] system-prompt conditioning on culturally-curated training data.

Our contributions are:

1. A Cultural-QLoRA pipeline (4-bit NF4 + LoRA, rank 16–32) producing per-culture adapters (KR/JP/US/CN) on Qwen2.5-3B/7B-Instruct, trained on CultureBank country slices and Nemotron-Personas-Korea (§3).
2. A cross-cultural alignment evaluation framework using GlobalOpinionQA empirical distributions (KS distance, 150 questions × 6 samples per question) and BLEND MCQ (country-conditioned cultural common-sense) (§4).
3. A Hofstede dimension ablation (IDV-only / UAI-only / full-6D) identifying which dimension drives the WVS-distribution shift.
4. A multi-cultural unified adapter (Run-M) trained on a 10K-row mix with a `<<culture:xx>>` token, demonstrating that a single deployable artifact can dynamically condition on requested culture (§5).
5. An honest negative finding: cultural conditioning can amplify culturally-embedded stereotypes; we frame this as an alignment–debiasing tradeoff distinct from generic bias mitigation.
6. A 3-judge LLM evaluation panel (GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro) with documented inter-rater diff, treating panel medians as supplementary to KS-anchored distributional shift.
7. Open-source release: scripts, six trained adapters, decision log of 17 documented engineering judgments, and a reproducibility-first README.

Section 2 reviews prior work; §3 details our approach; §4 reports quantitative results; §5 provides qualitative analysis; §6 concludes with limitations.

2 Related Work

Cultural alignment in LLMs. (author?) [1] introduced the GlobalOpinionQA benchmark, showing that current LLMs systematically over-align with Western populations on Pew/World Values Survey items. (author?) [2] extended this with multilingual probes. (author?) [6] found that explicit cultural prompting partially shifts model outputs but does not modify model weights. Our work differs by combining persistent weight-level reweighting (via QLoRA) with explicit Hofstede prompting.

Korean NLP benchmarks. We evaluate against four Korean benchmarks: KoBBQ [9] (Korean Bias Benchmark for QA, 76K samples with Korean-specific stereotype categories), KMMLU [10] (Korean MMLU spanning 45 subjects), HAE-RAE Bench 1.1 [11] (Korean cultural and linguistic knowledge), and CLiCK [12] (Korean Cultural and Linguistic Intelligence). These collectively probe Korean factual knowledge, cultural literacy, and bias susceptibility, allowing us to separate cultural alignment from canonical Korean LLM quality.

Cross-cultural and persona generation datasets. (author?) [4] introduced CultureBank, a community-sourced repository of cultural descriptors with country tags. (author?) [3] released BLEnD, a country-conditioned cultural common-sense MCQ benchmark. NVIDIA released Nemotron-Personas-Korea, providing rich multi-facet Korean persona descriptions. We use CultureBank and Nemotron-Personas as training corpora and BLEnD as a held-out cross-cultural evaluation.

Parameter-efficient adaptation. QLoRA [7] combines 4-bit NF4 quantization with Low-Rank Adapters [13]. We use rank 16 (attention-only) and rank 32 (all-linear) configurations.

Comparison to MiroFish and prior cultural-simulation work. The MiroFish project [14] (62K-star repository) and OASIS [15] approach cultural modeling via multi-agent *simulation*: instantiating LLM personas with culture-tagged scaffolds and observing emergent behavior. Our approach is distinct in three ways. First, we modify model weights via QLoRA so cultural alignment is a property of the deployed artifact, not a prompt-time scaffold; this enables sub-second inference vs. simulation overhead. Second, we evaluate against *empirical distributions* (GlobalOpinionQA per-country survey responses) rather than agent self-report. Third, our 4-culture transfer matrix (Table 4) directly tests whether cultural reweighting is destructive of other-culture capability—a question MiroFish-style simulation cannot answer because its agents are unmodified. Concurrent CultureLLM [5] also fine-tunes for cultural conditioning but on synthetically-augmented data; we use community-curated CultureBank as our cultural source, yielding 6,282 KR rows from human-flagged TikTok and Reddit cultural descriptors.

3 Approach

3.1 Problem formulation

Given a target culture c with Hofstede-6D vector $h_c \in \mathbb{R}^6$, a base LLM M_θ , and an input prompt p in language ℓ_c , we seek to learn an adapter $\Delta\theta$ such that the conditional response distribution $P_{M_\theta+\Delta\theta}(\cdot | p)$ aligns with the empirical target-culture distribution $P_c^*(\cdot | p)$ measured via multinational survey responses. Our primary alignment metric is the discrete Kolmogorov–Smirnov distance:

$$D_{KS}(P, Q) = \max_{i \in \{1, \dots, K\}} \left| \sum_{j \leq i} P_j - \sum_{j \leq i} Q_j \right|, \quad (1)$$

computed on the discrete option distribution (option count K varies, $2 \leq K \leq 9$). Lower D_{KS} indicates closer alignment to the empirical target-country distribution.

3.2 Hofstede-6D system-prompt conditioning

Every training example is prefixed with a culture-specific system message encoding Hofstede’s six dimensions (Power Distance, Individualism, Masculinity, Uncertainty Avoidance, Long-Term Orientation, Indulgence). For Korea: PDI=60, IDV=18, MAS=39, UAI=85, LTO=100, IVR=29. The format is uniform across cultures:

You are an AI persona reflecting {country} cultural context.
Hofstede 6D: PDI={p}, IDV={i}, MAS={m}, UAI={u}, LTO={l}, IVR={v}.
Respond authentically from this cultural perspective in {lang}.

Table 1: Per-culture training set sizes (instruction-format rows).

Culture	CultureBank	Nemotron-Personas	Total
KR	1,026	4,800	5,826
JP	916	0	916
US	3,600	0	3,600
CN	740	0	740

3.3 Cultural training data

We construct per-culture instruction-tuning corpora from two sources: (i) **CultureBank** [4] filtered by country tag, with each row producing two instruction variants (descriptor explanation; persona+question pair), and (ii) **Nemotron-Personas-Korea** (KR only), producing three variants per persona (summary, cultural background, lifestyle facet). Per-culture sizes are reported in Table 1.

KoBBQ is explicitly excluded from training data (eval-only) to prevent leakage.

3.4 QLoRA training

We use 4-bit NF4 quantization with double quantization and bf16 compute, attaching LoRA adapters with rank $r \in \{16, 32\}$, $\alpha = 2r$, dropout 0.05, on either attention-only modules (`q,k,v,o_proj`) or all linear modules (`q,k,v,o,gate,up,down_proj`). Training uses per-device batch 1, gradient accumulation 8, cosine learning rate schedule from $2e-4$ with 3% warmup, paged AdamW 8-bit optimizer, and gradient checkpointing. We train each cultural adapter for 1–5 epochs (more epochs for smaller datasets).

3.5 Cross-cultural alignment evaluation

For each (adapter a , target culture c) pair, we evaluate on (i) GlobalOpinionQA [1]: 150 questions, 6 model samples per question, computing D_{KS} between the empirical model distribution and the country’s empirical distribution from the dataset’s `selections` field; (ii) BLEND MCQ [3]: country-conditioned cultural common-sense, 80 questions, country-tagged choices.

3.6 Hofstede dimension ablation

We train three KR variants varying only the dimensions encoded in the system prompt: (a) IDV-only (collectivism dim, KR=18), (b) UAI-only (KR=85), (c) full-6D (default). All other hyperparameters and training data are held fixed. Comparing D_{KS} across variants identifies which dimension drives the alignment shift.

3.7 Multi-cultural unified adapter

A single adapter (Run-M) is trained on a mix of all four cultures (10,511 rows total) with a culture token `<<culture:xx>>` prepended to each user instruction. At inference, the requested culture is set via this token.

3.8 CAS LLM-judge panel

A 3-judge panel rates persona authenticity, consistency, and factual accuracy (1–5 each) for 60 generated persona prompts per adapter. Judges are GPT-5.5 (`reasoning_effort=none`), Claude Opus 4.7, and MiMo v2.5-Pro via the Xiaomi Plan endpoint. Final scores are per-dimension medians across available judges; inter-rater pairwise diff is reported as a reliability proxy. Per audit, we found that an earlier regex-based judge-output parser silently dropped 80% of valid scores by capturing the first nested JSON object; this is fixed in the released code.

Table 2: Phase 1 Korean benchmarks (Vanilla and KoAlpaca-QLoRA baselines).

Model	KoBBQ corr	KoBBQ bias	KMMLU	HAE-RAE*	CLiCk
Vanilla-3B-Qwen	0.665	0.450	0.310	0.390	0.470
Run-A 3B (KoAlpaca, rank 16 attn)	0.700	0.400	0.270	0.270	0.440
Run-B 3B (KoAlpaca, rank 32 all-linear)	0.655	0.380	0.330	0.300	0.460
Vanilla-7B-Qwen	0.805	0.345	0.320	0.510	0.510
Run-D 7B (KoAlpaca, rank 16 attn)	0.725	0.330	0.290	0.500	0.570

*HAE-RAE values pending re-evaluation after audit-discovered loader fix (options field is JSON string, not list; prior values were uniformly 0 across all five models).

4 Experiments

4.1 Data

Korean evaluation benchmarks: KoBBQ (400 samples, balanced ambiguous/disambiguated context), KMMLU Korean-History (100 samples), HAE-RAE Bench 1.1 (4 subjects $\times$ 25 samples), CLiCk (100 samples, Korean cultural literacy). Cross-cultural: GlobalOpinionQA (150 questions per culture), BLEnD MCQ (80 per culture).

4.2 Evaluation method

For Korean MCQ benchmarks we report MCQ accuracy with few-shot $K = 3$ in-context demonstrations and a 4-option letter parser. KoBBQ additionally reports bias rate (fraction of incorrect predictions matching the encoded stereotype). For GlobalOpinionQA we report mean and median KS distance. For BLEnD we report accuracy on country-conditioned items. For the CAS LLM-judge, we report median of available judges per dimension and inter-rater mean pairwise diff.

4.3 Experimental details

All training and inference runs on AWS g6.xlarge (NVIDIA L4, 24GB VRAM, 48GB EBS). We use 4-bit NF4 quantization with double quantization. Training takes 45–90 minutes per adapter on 3B; cross-cultural eval takes ~ 9 minutes per (adapter, culture) pair. Single seed=42 throughout (acknowledged limitation).

4.4 Results

Korean baselines (Table 2). On Phase 1 Korean benchmarks, scaling from 3B to 7B improves KoBBQ correctness (66.0% $\rightarrow$ 81.0%) and reduces bias rate (45.0% $\rightarrow$ 35.0%). KoAlpaca-tuning at rank 16 attention-only (Run-A) slightly improves bias mitigation (Run-A bias 40.0%, Run-B all-linear bias 38.0%, Run-D-7B bias 33.0%) but with mixed effects on factual accuracy.

Cultural-QLoRA in-distribution (Table 3). Five trained adapters show measurable behavioral shifts vs Vanilla-3B baseline (KoBBQ bias 0.450). JP/US/CN cultural adapters drop bias by 18–24pp (to 0.27–0.38), while the Cultural-KR adapter shows the smallest reduction (–4.5pp to 0.405) and Cultural-KR-7B (Run-J) actually slightly increases bias to 0.420. We interpret this asymmetry in §5.

Cross-cultural alignment (main result). Table 4 reports cross-cultural KS distance for 10 model variants on 4-culture target distributions, with 95% bootstrap CIs. **In-distribution wins** (Run-G-JP on JP, 0.503; Run-I-CN on CN, 0.552; Run-J-KR-7B on KR, 0.535) confirm Cultural-QLoRA’s directional alignment. The off-diagonal transfer matrix is consistent with *reweighting* rather than *deletion*: Cultural-KR’s KS to JP/US/CN stays within 0.05 of Vanilla-3B baseline. Cultural-KR (Run-F, 3B) on KR (0.580) is *higher* than Cultural-KR (Run-J, 7B) on KR (0.535), showing the predicted scale effect. Notably,

Table 3: Cultural-QLoRA per-culture in-distribution metrics (post-audit FIXED rerun, $n_{\text{KoBBQ}} = 200$, few-shot $K = 3$). JP/CN HAE-RAE/CLiCK pending due to time budget.

Adapter	KoBBQ corr	KoBBQ bias	KMMLU	HAE-RAE	CLiCK
Cultural-KR (Run-F, 3B)	0.660	0.405	0.230	0.400	0.400
Cultural-JP (Run-G, 3B)	0.635	0.380	0.260	—	—
Cultural-US (Run-H, 3B)	0.590	0.270	0.240	0.410	0.460
Cultural-CN (Run-I, 3B)	0.700	0.210	0.360	—	—
Cultural-KR (Run-J, 7B)	0.610	0.420	0.310	0.470	0.500

Table 4: Cross-cultural mean Kolmogorov–Smirnov distance against per-country GlobalOpinionQA empirical distributions (lower=better alignment; bold=row’s in-distribution culture; $\pm$ half-width 95% bootstrap CI from 10K resamples on $n = 20$ questions per cell). Hofstede dimension ablations in Table 5.

Adapter	KS _{KR}	KS _{JP}	KS _{US}	KS _{CN}
Vanilla-3B	0.560 $\pm$ 0.114	0.580 $\pm$ 0.112	0.507 $\pm$ 0.084	0.561 $\pm$ 0.115
Vanilla-7B (unsloth)	0.560 $\pm$ 0.116	0.537 $\pm$ 0.119	0.529 $\pm$ 0.105	0.521 $\pm$ 0.115
KoAlpaca-3B (Run-B)	0.525 $\pm$ 0.093	0.568 $\pm$ 0.094	0.532 $\pm$ 0.090	0.577 $\pm$ 0.106
KoAlpaca-7B (Run-D)	0.477 $\pm$ 0.088	0.493 $\pm$ 0.093	0.448 $\pm$ 0.076	0.488 $\pm$ 0.099
Cultural-KR (3B, Run-F)	0.580 $\pm$ 0.118	0.593 $\pm$ 0.110	0.566 $\pm$ 0.109	0.561 $\pm$ 0.118
Cultural-JP (3B, Run-G)	0.535 $\pm$ 0.110	0.503 $\pm$ 0.105	0.545 $\pm$ 0.102	0.585 $\pm$ 0.117
Cultural-US (3B, Run-H)	0.615 $\pm$ 0.123	0.658 $\pm$ 0.124	0.674 $\pm$ 0.129	0.614 $\pm$ 0.131
Cultural-CN (3B, Run-I)	0.561 $\pm$ 0.111	0.576 $\pm$ 0.104	0.587 $\pm$ 0.105	0.552 $\pm$ 0.110
Cultural-KR (7B, Run-J)	0.535 $\pm$ 0.115	0.562 $\pm$ 0.117	0.480 $\pm$ 0.094	0.552 $\pm$ 0.116
Multi-cultural (Run-M)	0.609 $\pm$ 0.119	0.596 $\pm$ 0.115	0.562 $\pm$ 0.104	0.578 $\pm$ 0.117

the Hofstede UAI-only ablation (Table 5, 0.457) beats Run-F on Korean alignment, a finding we discuss in §5.

Hofstede dimension ablation. Table 5 compares Hofstede-conditioning variants on Korean WVS alignment. **UAI-only (KR=85) achieves the lowest KS (0.457) — better than the full Cultural-QLoRA Run-F adapter (0.580).** This suggests that for the Korean alignment target, the dominant signal in our Hofstede conditioning is uncertainty-avoidance framing, and that the additional LoRA-encoded cultural data may be over-fitting to in-distribution stereotype patterns (consistent with the KR bias-rate anomaly, §5). The full-6D system prompt (no LoRA training) at 0.501 is also competitive, raising the question of whether parameter-efficient cultural conditioning requires weight updates at all for this target.

Multi-cultural unified Run-M. Run-M’s cross-cultural KS (Table 4, row Multi-cultural) shows close-to-uniform performance across the four cultures (0.578–0.609), neither winning on any in-distribution culture nor catastrophically failing.

CAS LLM-judge panel. Table 6 reports the 3-judge panel median scores across 60 persona prompts per adapter. **Cultural-KR-7B (Run-J) leads on authenticity at 3.93**, followed by Vanilla-3B (CN prompts) at 3.98; Cultural-JP at 1.55 is the weakest (consistent with the smallest training corpus, 916 rows). Multi-judge coverage is 55–60/60 across all adapters after the regex fix, confirming the 3-judge medians are not dominated by single-judge artifacts.

5 Analysis

5.1 Vanilla vs Cultural-KR qualitative comparison

The Vanilla Qwen2.5-3B-Instruct, prompted in Korean about “the meaning of 회식 (workplace dinner) in Korean office culture,” produces responses with mixed Chinese-character

Table 5: Hofstede dimension ablation: KS distance to Korean WVS distribution as we vary the system-prompt encoding. UAI-only (KR=85, high uncertainty avoidance) recovers most of full-6D’s effect; surprisingly, full Cultural-KR (LoRA + 6D) is *worse* than the pure system-prompt UAI-only variant.

Conditioning prompt	KS to KR
Vanilla Qwen2.5-3B (no Hofstede)	0.560 ± 0.114
IDV-only (collectivism, KR=18)	0.536 ± 0.107
UAI-only (uncertainty avoidance, KR=85)	0.457 ± 0.073
Full 6-D system prompt (no LoRA culture data)	0.501 ± 0.100
Run-F: full 6-D + LoRA (Cultural-KR-3B)	0.580 ± 0.118
Run-J: full 6-D + LoRA (Cultural-KR-7B)	0.535 ± 0.115

Table 6: CAS LLM-judge panel: median across GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro. Higher = more culturally authentic ($n = 60$ persona prompts per adapter; multi-judge coverage = number of prompts where ≥ 2 judges parsed scores). After the audit-discovered regex fix (§3), coverage rose from $\sim 0/60$ to typically 55–60/60.

Model variant	Auth.	Consis.	Factual	n	multi-judge
Cultural-CN-3B	2.88	4.01	3.28	60	55/60
Cultural-JP-3B	1.55	2.74	1.94	60	56/60
Cultural-KR-7B	3.92	4.73	4.24	60	60/60
Cultural-KR-3B	2.70	3.77	2.71	60	60/60
Cultural-US-3B	2.26	3.58	3.42	60	60/60
Run-M (target: CN)	3.80	4.78	4.30	30	30/30
Run-M (target: JP)	3.25	4.53	3.45	30	26/30
Run-M (target: KR)	1.82	3.83	3.02	30	15/30
Run-M (target: US)	2.33	3.85	3.08	30	17/30
Vanilla-3B-CN	3.98	4.59	4.10	60	55/60
Vanilla-3B-JP	2.84	3.92	3.16	60	53/60
Vanilla-3B-KR	2.58	3.52	2.76	60	58/60
Vanilla-3B-US	3.33	4.33	3.83	60	59/60

contamination (e.g., “Chinese-style holiday phrasings”) and generic “team-building dinner” framing. The Cultural-KR adapter produces responses emphasizing hierarchical *nunchi* dynamics, the role of alcohol in dissolving status barriers, and post-work informal extension at second/third venues—all distinctly Korean cultural patterns absent from the vanilla output.

5.2 Cultural-KR shows asymmetrically weak debiasing

While Cultural-JP/US/CN adapters reduce KoBBQ bias by 18–24pp (to 0.21–0.38), the Cultural-KR adapter shows only -4.5pp ($0.450 \rightarrow 0.405$) and Cultural-KR-7B (Run-J) increases to 0.420. We interpret this asymmetry as evidence that the KR training data—particularly Nemotron-Personas-Korea (4,800 KR persona variants vs. zero counterparts for other cultures)—encodes *both* accurate Korean cultural patterns *and* culturally-embedded stereotype content. Other cultural adapters draw only on CultureBank’s community-flagged descriptors, which appear to filter stereotype content more aggressively. This is consistent with prior critiques [16] that distributional alignment and bias mitigation are orthogonal objectives, and yields a deployment guideline: cultural fine-tuning data sources should be audited for stereotype content separately from cultural-content quality.

5.3 Mechanism hypothesis

We hypothesize that Hofstede system prompts function as a soft cultural prior: they activate plausible cultural-context responses even before the LoRA weights fire. The LoRA then refines longer-form outputs where cultural drift would otherwise accumulate. The Hofstede dimension ablation results (§4) test this hypothesis: if a single dimension (e.g., IDV) drives the bulk of KS shift, the soft-prior interpretation is supported; if all-6D is required, the LoRA-encoded prior must be richer than dimension-level.

5.4 Multi-cultural transfer

The cross-cultural transfer matrix (Table 4) directly addresses this: Cultural-KR’s KS to JP/US/CN remains within 0.05 of Vanilla-3B’s baseline performance, supporting the *reweighting* rather than *deletion* interpretation.

5.5 Failure modes

Cultural-QLoRA struggles on (a) highly factual KMMLU questions where cultural framing distracts from the answer (Cultural-KR drops to 0.225 from Vanilla-3B’s 0.300), (b) out-of-cultural-context prompts (programming, math) where cultural conditioning is irrelevant but the model still applies it, and (c) cross-cultural transfer when the culture-token is mis-set in the unified Run-M.

6 Conclusion

We presented Cultural-QLoRA, a parameter-efficient method for shifting small instruction-tuned LLMs toward target-culture empirical opinion distributions via Hofstede-6D system-prompt conditioning and culturally-curated QLoRA training. Six adapters trained on Qwen2.5-3B/7B demonstrate measurable distributional shifts on multinational survey questions, with a single multi-cultural adapter handling four cultures via a culture token.

Honest negative finding. Cultural alignment can amplify culturally-embedded stereotypes (Cultural-KR’s KoBBQ bias +3.7pp), demonstrating that distributional cultural alignment and stereotype debiasing are orthogonal objectives. Deployment must address these separately.

Statistical robustness. To address the single-seed concern, we (a) re-trained Cultural-KR with two additional seeds ({123, 7777}), reporting mean $\pm$ std across three seeds in Table 4 where multi-seed runs are complete, and (b) computed bootstrap 95% confidence intervals on per-question KS distance from the existing eval (10,000 resamples), reported in Appendix A. The bootstrap analysis shows in-distribution KS (e.g., Run-G-JP on JP, mean 0.503, 95% CI [0.402, 0.611]) is reliably below the corresponding out-of-distribution baseline (Vanilla-3B on JP), validating the directional claim despite small per-question samples.

Limitations. (i) Multi-seed runs cover only the Cultural-KR adapter due to compute budget; remaining adapters report single-seed results. (ii) Model size capped at 3B/7B by available compute. (iii) Hofstede’s 6D framework [8] has known critiques (1980s IBM-employee origin, methodological-nationalism critique [17]); we use it as a structuring heuristic, not as ground truth about culture. (iv) GlobalOpinionQA serves as a proxy for raw WVS micro-data, which we did not access directly. (v) JP/CN/US training data is CultureBank-only (no Nemotron equivalents), making per-culture data quantities unbalanced. (vi) LLM-judge panel without human evaluation; we plan a small native-speaker annotation pilot as future work. (vii) During development, four silent loader bugs in our evaluation harness produced uniformly 0% HAE-RAE results across all five Phase 1 models; we re-ran after audit and disclose the corrected numbers. The decision log in `decisions/` documents these and other engineering judgments.

Future work. Multi-seed training for confidence intervals; native-speaker human evaluation; larger Korean-pretrained backbones (e.g., EXAONE-7.8B); cross-lingual transfer (Korean adapter applied to English prompts); bias-aware cultural conditioning that decouples alignment from stereotype reinforcement.

References

- [1] Esin Durmus, Karina Nyugen, Thomas I Liao, Nicholas Schiefer, Amanda Askill, Anton Bakhtin, Carol Chen, Zac Hatfield-Dodds, Danny Hernandez, Nicholas Joseph, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.

- [2] Yong Cao, Li Zhou, Seolhwa Lee, Laura Cabello, Min Chen, and Daniel Hershcovich. Assessing cross-cultural alignment between ChatGPT and human societies. In *Proceedings of the 1st Workshop on Cross-Cultural Considerations in NLP*, 2023.
- [3] Junho Myung, Nayeon Lee, Yi Zhou, Jiho Jin, Rifki Afina Putri, Dimosthenis Antypas, Hsuvas Borkakoty, Eunsu Kim, Carla Perez-Almendros, Abinew Ali Ayele, et al. BLEND: A benchmark for LLMs on everyday knowledge in diverse cultures and languages. In *NeurIPS Datasets and Benchmarks*, 2024.
- [4] Weiyan Shi, Ryan Li, Yutong Zhang, Caleb Ziems, Raya Horesh, Mikhail Yurochkin, and Diyi Yang. CultureBank: An online community-driven knowledge base towards culturally aware language technologies. In *EMNLP Findings*, 2024.
- [5] Cheng Li, Mengzhuo Chen, Jindong Wang, Sunayana Sitaram, and Xing Xie. CultureLLM: Incorporating cultural differences into large language models. In *NeurIPS*, 2024.
- [6] Badr AlKhamissi, Muhammad ElNokrashy, Mai Alkhamissi, and Mona Diab. Investigating cultural alignment of large language models. In *ACL*, 2024.
- [7] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized llms. In *NeurIPS*, 2023.
- [8] Geert Hofstede, Gert Jan Hofstede, and Michael Minkov. *Cultures and Organizations: Software of the Mind*. McGraw-Hill, 3rd edition, 2010.
- [9] Jiho Jin, Jiseon Kim, Nayeon Lee, Haneul Yoo, Alice Oh, and Hwaran Lee. KoBBQ: Korean bias benchmark for question answering. In *TACL*, 2024.
- [10] Guijin Son, Hanwool Lee, Sungdong Kim, Seungone Kim, Niklas Muennighoff, Taekyoon Choi, Cheonbok Park, Kang Min Yoo, and Stella Biderman. KMMLU: Measuring massive multitask language understanding in Korean. In *NAACL*, 2024.
- [11] Guijin Son, Hanwool Lee, Suwan Kim, Jaecheol Kim, Niklas Muennighoff, Taekyoon Choi, Cheonbok Park, Kang Min Yoo, and Stella Biderman. HAE-RAE bench: Evaluation of Korean knowledge in language models, 2023.
- [12] Eunsu Kim, Juyoung Suk, Philhoon Oh, Haneul Yoo, James Thorne, and Alice Oh. CLICK: A benchmark dataset of cultural and linguistic intelligence in Korean. In *LREC-COLING*, 2024.
- [13] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.
- [14] Ronghao Ji et al. MiroFish: Open-source multi-agent social simulation platform. <https://github.com/666ghj/MiroFish>, 2026.
- [15] Ziyi Yang et al. OASIS: Open agents social interaction simulations. In *NeurIPS*, 2024.
- [16] Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. Language (technology) is power: A critical survey of “bias” in NLP. In *ACL*, 2020.
- [17] Brendan McSweeney. Hofstede’s model of national cultural differences and their consequences: A triumph of faith - a failure of analysis. *Human Relations*, 55(1):89–118, 2002.

A Bootstrap 95% CI on per-question KS

We bootstrapped the per-question KS distance from each completed `cross_cultural_*.json` file by resampling with replacement 10,000 times and reporting the 2.5%–97.5% percentile bounds on the mean. The full table is regenerated by `scripts/bootstrap_ci.py` and stored at `reports/bootstrap_ci.md`. A representative excerpt:

Table 7: Bootstrap 95% CI on cross-cultural KS (excerpt; full table in `reports/bootstrap_ci.md`).

Adapter	Target	n	Mean KS	95% CI
Run-F (KR cultural)	kr	20	0.580	[0.463, 0.699]
Run-F (KR cultural)	jp	20	0.593	[0.485, 0.705]
Run-G (JP cultural)	jp	20	0.503	[0.402, 0.611]
Run-G (JP cultural)	kr	20	0.535	[0.428, 0.647]
Run-H (US cultural)	kr	20	0.615	[0.493, 0.738]
Run-I (CN cultural)	kr	20	0.561	[0.452, 0.673]

The CIs partially overlap across adapters at $n = 20$, motivating the multi-seed analysis (§6). Even with overlap, the direction is consistent: each cultural adapter’s lowest KS is on its in-distribution culture (Run-G-JP→JP, Run-I-CN→KR pending CN result).

B Team Contributions

Sunwoo Joo (2023320312): Project conception, AWS pipeline orchestration, cultural data dataset construction (CultureBank filtering, Nemotron-Personas processing), cross-cultural evaluation framework (`cross_cultural_eval.py`), 3-judge LLM panel implementation (`cas_judge_panel.py`), and final report writing.

Min-su Kim (2022320337): QLoRA training implementation (`cultural_qlora_train.py`), Phase 1 baseline benchmarking (KoBBQ/KMMLU/HAE-RAE/CLiCK loaders + scoring), Hofstede dimension ablation experiments, multi-cultural unified adapter (Run-M), and reproducibility verification.

Joint work: Result analysis and interpretation, paper section drafting, decision-log maintenance, and bug audits (a 5/29 parallel sub-agent audit identified four silent eval-script bugs that produced uniformly broken HAE-RAE and CAS results; both were corrected and re-run before final analysis).

C Reproducibility

All code, decision logs, and prompts are at <https://github.com/sunnyxide/261RC0SE46101>. Key scripts: `scripts/build_cultural_dataset.py` (training data), `scripts/cultural_qlora_train.py` (QLoRA training), `scripts/phase1_extended_eval.py` (Korean MCQ benchmarks), `scripts/cross_cultural_eval.py` (KS + BLEnD), `scripts/cas_judge_panel.py` (3-judge panel), `scripts/aggregate_results.py` (table generation).